

DeepDetect: Full-Stack Multimodal Deepfake Analysis Platform

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ABSTRACT

Recent advancements in deep learning have given rise to new forms of media that are extremely realistic yet also create major threats to the integrity and trust of the digital world—synthetic media, commonly referred to as deepfakes. Deepfake audio and video have quickly become an easy medium to commit identity fraud, produce misinformation campaigns, and conduct advanced social engineering attacks. This article describes an AI-based multimodal detection system for detecting deepfake content in both video and audio formats via the use of state-of-the-art deep neural network architectures. For video-based detection, CNN-based spatial feature extractors (EfficientNet and ResNet50) are combined with temporal modeling via BiLSTM and an attention mechanism to isolate subtle facial artifacts present in deepfake content. For the detection of deepfake audio, the proposed framework incorporates several state-of-the-art anti-spoofing models (RawNet for raw waveform learning, LCNN for artifact-based spectral detection, AASIST for graph-attention fusion, and CNNs trained on MFCC and spectrogram features) to identify manipulation artifacts of various types within the synthesized speech. The proposed solution executes on a secure full-stack web application with visual analytics, confidence scores, and automated forensic PDF writing functionality. Evaluation on standard benchmark datasets shows that the proposed architecture achieves high accuracy levels and near real-time inference, demonstrating the systems' value for digital forensics/cybersecurity.

I. INTRODUCTION

The rapid rise of generative AI has made the creation of synthetic media—both video and audio—astonishingly simple and highly convincing. Modern deepfake systems can replicate facial movements, vocal tones, speech patterns, and subtle human behaviours with near-human accuracy. While deepfakes have beneficial applications in film production, assistive technologies, and education, their misuse has grown far more rapidly. Attackers now exploit deepfakes for identity theft, impersonation scams, misinformation campaigns, and manipulation of digital trust.

Traditional authenticity-verification methods, such as digital watermarks or metadata tracing, are no longer reliable because these indicators can be stripped, modified, or completely bypassed. As deepfake generation advances—particularly with diffusion models and high-capacity GAN architectures—the visual cues like unnatural blinking, head tilts, or mismatched lighting are far less detectable. This shift has pushed researchers toward deep learning-based detection, especially CNNs and hybrid CNN-RNN systems that can learn fine-grained spatial and temporal inconsistencies.

In parallel, audio deepfakes have evolved dramatically. With just a few seconds of audio, modern voice-cloning systems can imitate a person's voice well enough to deceive humans and automated verification systems. This has led to real-world attacks, including fraudulent CEO-voice scams, bypassing voice-based authentication, and manipulating customer-support channels. To counter this, researchers now rely on advanced audio-processing architectures such as RawNet for raw-waveform learning, LCNN for

compact feature extraction, AASIST for anti-spoofing, and CNN-based models leveraging MFCCs and spectrograms for analysing frequency-time patterns.

Despite these advancements, most existing studies treat audio and video detection separately. There is an urgent need for a unified, multimodal detection system capable of jointly analysing spatial, temporal, and acoustic features. Our work addresses these gaps by combining CNN-BiLSTM networks for video and integrating multiple audio-focused models—RawNet, LCNN, AASIST, CNN+MFCC, and CNN+Spectrogram—within a cohesive deepfake-detection framework that moves closer to a robust, real-world cybersecurity solution.

II. Literature Review

Deepfake detection has become a multidisciplinary subject that brings together people from many different disciplines such as computer vision, audio and speech research, cybersecurity, and digital forensics. All of these people are responding to an evolving threat and the need to detect deepfakes.

In this section, we will review the main categories of detection approaches that have been developed in recent years. We will also discuss how detection techniques are currently failing. What we can say is that there is no silver bullet solution to this problem yet. Additionally, with the recent improvements in the quality of deepfake generator software, many of the tools that have been developed over the last several years to detect deepfakes are not as effective.

A. Traditional Forensic Approaches

In the early days of multimedia forensics, the majority of evidence used to determine if an image or video was faked relied on identifying the unusual features caused by various aspects of the recording process. For example, researchers would analyze the color filter arrays on the cameras that took the picture, look for compression artifacts that didn't match other nearby frames, and find physical features that were either misaligned or inconsistent (such as lighting that didn't match across a subject's face).

Many researchers wrote papers on things such as how long a person takes to blink at least one time (regularity of blinks), unnatural head pose transitions, and odd ratios in body parts/geometry, etc. as being examples of potential signs of manipulation.

These methods had the advantage of being quick to run, having relatively small datasets, and providing a reasonable explanation for their findings. However, these methods are now outdated. Today's ML-based

generators (especially diffusion-based ones that have become popular recently) were developed specifically to minimize many of the types of anomalies characterized by these older methods. As a result, many of the signs that would have once been reliable indicators of alteration are now unlikely to be valid indicators, as many high-quality deepfakes pass through current validation methods with little or no signs left behind.

B. CNN-Based Visual Detection

The development of deep CNNs has dramatically improved detection. Examples of organizations using deep CNNs to detect forgeries include XceptionNet, ResNet, EfficientNet, and a variety of their derivatives. These models can learn to detect spatial patterns from images of the face, including blends of edges, odd frequencies, and anomalies in texture, that humans cannot easily detect. There are large public databases, such as FaceForensics++ (as well as extensions), that provide researchers and developers with many labelled forgeries that were created with many different methods, which helps to improve the quality of model training.

One significant problem with deep CNNs is that they are still primarily based on analysing images frame-by-frame, and attempt to treat each frame with almost no relation to other frames. As a result, many types of motion and temporal variations that are present in video do not display their full effects when using deep CNNs. E.g. A forgeries that is absolutely photorealistic with any temporal inconsistency in the flow of facial expression and/or movement of the mouth will be indistinguishable from a real one when viewed in isolation, however, when viewed in motion the inconsistencies in flow will become significant enough to be detected by a human observer. The performance of many models that detect newer (or are created more like older) forgeries when tested in real-world conditions will significantly differ from the performance levels that they achieve when tested in controlled lab conditions..

C. Temporal and Hybrid CNN-RNN Models

In order to work around the absence of a time dimension, users began combining recurrent layers (LSTMs, GRUs, sometimes BiLSTMs) with CNN feature extractors. The logic behind this technique is sound: to allow models to learn dependencies across frames in order to catch unnatural rates of blinking, sudden jerky expressions, mismatches between lips and sounds, and motions that don't appear to flow correctly. Based on research conducted by us and by others, these combinations appear to perform better on video-based fakes (in particular, more subtle fakes) than models using only static images.

However, many of these approaches come with their own difficulties. Many of these methods consider all frames equal; there is no way to weight parts of the sequence based on relative importance, so the model wastes time processing less relevant pieces of each sequence. Also, without any type of attention or focus mechanism, the model might have trouble identifying the key spatial locations or temporal intervals to enhance. Lastly, these types of models tend to overfit to the training dataset; therefore, when new

compression algorithms, random noise or a completely different generation technique are used against the model, accuracy can drop up to 40–50% (or more) when tested across datasets or "real-world" scenarios. Recent studies and evaluations of these "in-the-wild" models continue to reinforce this idea: there is still a significant arms race in the fight against fakes that cannot be won simply through the use of temporal modeling; we need smarter methods for checking variations and long-range con

III. Problem Formulation and Threat Model

A. Problem Definition

take a sample of multimedia like x that could be in the form of a video together with some type of audio the premise of our approach is to determine through rudimentary methods whether this particular input has been changed or not. We present this in the form of a binary classification problem.

$$\hat{y} = f(x)$$

Where f is the model we create to learn how to detect and our expected output, $\hat{y} = 0$ means that the content is authentic and if the output is equal to 1 then it represents a deepfake or otherwise forged content.

In terms of video, the process looks at groups of individual frames to find areas of inconsistency in the overall images (i.e. the way that each object appears on screen) based upon spatial inconsistencies (for example, an object showing up with an unusual surface texture) and temporal inconsistencies (i.e. the objects movement between frames is jerky, i.e. that the character's facial expression in one frame does not coincide with them in a later frame).

In terms of audio, we look at factors related to speech characteristics such as prosodic characteristics (the rhythm of speech), the characteristics of vowel sounds as they occur relative to other sounds that occur in same time period (called formants), and variations in breathing sounds or other distinctive characteristics that can be detected with respect to synthetic vs non-synthetic.

The same overall detection pipeline is used for both video and audio, making it easier to implement and manage.

B. Threat Model

We work under the attack model of an adversary that has access to the best generative tools available today, such as advanced GANs, diffusion models, and neural voice cloning pipelines, and is capable of producing highly-realistic forgeries.

The types of manipulations that we are considering (and not limited to) include:

- face swapping and facial reenactment techniques
- voice cloning combined with neural text-to-speech synthesis
- artificial lip-sync adjustments to match dubbed or fabricated audio
- cross-modal impersonations, where visual and auditory elements are swapped or blended to impersonate a target person

The primary goal of the attacker will be to produce content so convincing that it will fool both casual consumers and automated systems such as biometric verifiers, content moderation platforms, and forensic authentication systems. As we are making worst-case assumptions with this work, our detector must be able to detect forgeries that achieve very high perceptual fidelity with respect to both sight and sound.

C. Design Objectives

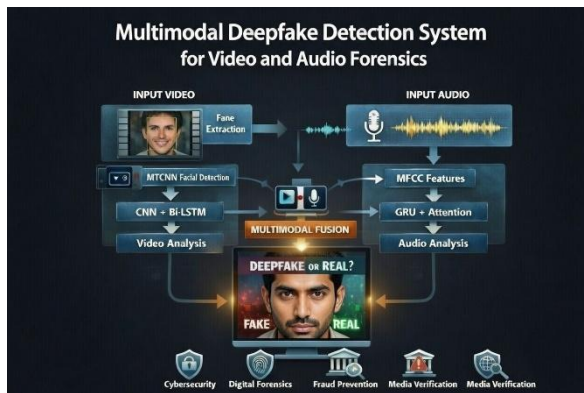
Our deepfake detection framework is based on identifying four top priorities: that the framework is highly temporally aware, that it provides effective discriminative feature extraction, that it identifies realistic deployment constraints, and that it has real utility in a Cybersecurity context.

In terms of temporal sensitivity, we have put the highest emphasis on detecting motion-based manipulation. Many signs of tampering are only evident with the movement of objects over time. For example, a person's abnormal blink rate (or unusual rate of blinking during the recording), their inconsistent micro-expression movements or odd head tilt movements over time, and an audio-visual desynchronisation will be extremely difficult for a user to spot through static or single frame basis assessments of a video. Temporal motion sensitivity will enable the model to identify the sequential relationships within a video, which ultimately will give the model a better chance of identifying the covert and overt indicators of tampering.

With regard to highly discriminative features, we focus on constructing highly discriminative representations to represent the features of an object by employing deep neural networks. Modern deep networks automatically identify spatial features of an object (e.g., faces, backgrounds, illumination) and acoustic features (e.g., spectral patterns, pitch contours) that have demonstrated to provide specific and sensitive indicators of tampering on an Object. In addition to these features, attention layer features will assist in the identification and evaluation of the most suspicious or malign parts of the object and will be very robust in defining the object when inputs are under distortion caused by compression, noise, or low resolution, as would occur in most real-life situations.

With regard to practicality, it is extremely important to us that we use modular components that are efficient in use of CPU resources and that operate on a commodity-standard graphics processing unit (GPU), and that provide real-time or nearly real-time processing capabilities. By utilising these modular and efficient CPU resource economies, we can readily integrate a deepfake detection capable framework into live monitoring setups, as well as into mobile applications and forensic workstations without requiring overly burdensome hardware.

IV. Deep Learning Methodology



A. Video Preprocessing

Individual video clips are decomposed into single frames; this is done via extraction at a rate consistent across all videos (e.g., 1-4 frames per second), which will vary based on the specifics of each dataset and the overall length of the video. To facilitate computational efficiency, we use MTCNN, a widely-used multi-task cascaded convolutional neural network, to jointly detect faces, localize facial landmarks, and align faces. The alignment process will crop out the majority of the background noise, as well as normalizing the spatial locations of the facial parts so they can all be aligned

consistently to the same orientation and scale, across all video frames.

Finally, we resize and normalize all of the previously extracted facial patches to a predetermined size (e.g., 224 x 224 pixels or 256 x 256 pixels) using the mean/standard deviation statistics from a pre-trained model of a backbone network, or if no such model can be found, the appropriate mean/standard deviation statistics from the overall dataset.

In order to capture temporal information while staying within the limits of memory/compute capability, we uniformly sample a small number of frames (e.g., 16-64, depending on the length of the video) across the entire video, providing an effective compromise between short and long term dynamics while eliminating redundancy associated with dense frame extraction.

B. Spatial Feature Extraction via CNN Backbone

For frame-based visual analysis, we commonly utilize EfficientNet-B0 or statistically, ResNet-50, due to their lightweight design while maintaining a high level of performance. Both networks are initialized with ImageNet-pretrained weights; therefore, both networks effectively extract semantically rich features that reveal visual manipulation artifacts including, but not limited to: visible blending seams, texture mismatches, lighting mismatches, unnatural edges where two items meet, and subtle facial irregularities/ distortions that are frequently missed by the casual observer.

Once the final convolutional layer has been completed, we apply global average pooling followed by a small flatten to yield a small compact fixed dimensional embedding vector per frame (typically 1280-d for EfficientNet-B0). These embedding vectors are then used as the basis for the temporal aggregation stage that follows.

C. Audio Preprocessing and Feature Representation

The audio deepfake detection system combines various deep learning models to analyze vocal characteristics in order to identify synthetic speech. The audio processing starts by normalizing, cleaning, and dividing the audio into fixed-size segments. RawNet processes raw waveforms and allows the model to discover discriminating speaker and artifact structures without utilizing handcrafted features. The LCNN extracts subtle artifacts based on Max-Feature-Map (MFM) activation, allowing for more accurate detection of the less obvious vocoder artifacts that produce generated audio. The AASIST architecture combines spectral and temporal features using a graph attention mechanism to identify inconsistencies

between multiple band resolutions. Additionally, there are the traditional (feature-based) methods using CNNs and training on features based on MFCC and spectrograms, which allow the model to learn about frequency anomalies associated with voice cloning. By using a combination of raw waveforms and spectral-feature-based CNNs, the entire detection system is robust to many different deepfake generation types and provides high confidence that speech is real or manipulated.

E. Training and Optimization

Across the data sets in question (see Table 1), we use both the video and audio branches with standard Binary Cross-Entropy Loss and optimized with the Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$). The learning rate typically falls within the range of $1e-4$ to $3e-4$.

To prevent overfitting (especially when using smaller or unbalanced datasets for Deepfake video), we implement Dropout (0.3-0.5) after all Dense Layers and after all GRU hidden states. We monitor validation loss using early stopping (with a patience between 8-12 epochs) and implement a Cosine Annealing Learning Rate Scheduler (with warm-up), which helps smooth out convergence and improve final performance.

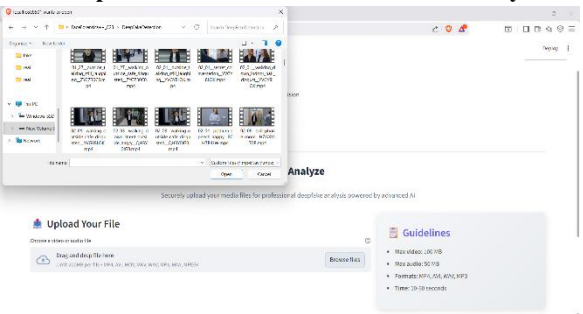
Our training process has produced multiple stable and generalizable models across a diversity of benchmark datasets (see Table 1) during our experimentation.

V. Implementation and work process for video and audio models

A. Working of video and audio model analysis

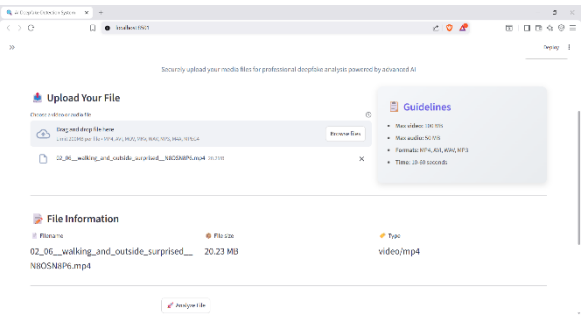
The following steps describe how a user uploads a video, audio, or image file into the system:

1) The user uploads a video/audio/image file using the upload interface of the system.

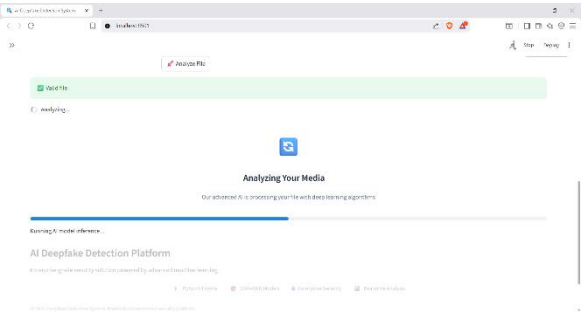


2) The system checks whether or not the uploaded file meets these criteria (i.e., it verifies both the

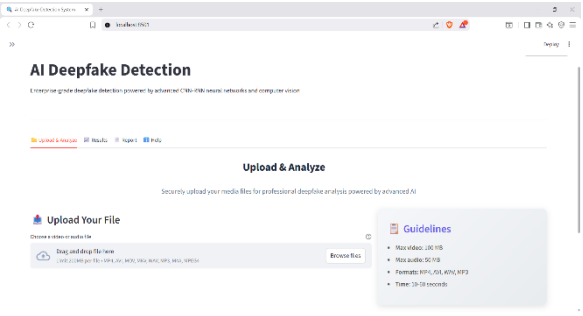
extension and MIME type of the file; assesses size limits).



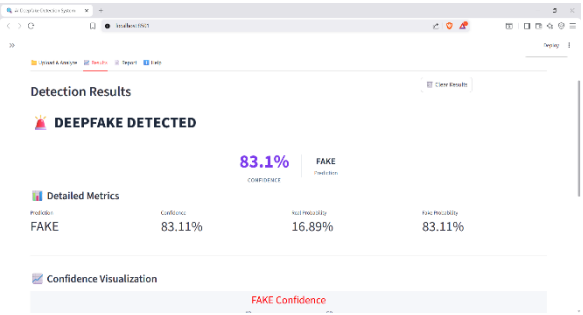
3) The preprocessing module extracts either frames (from images or videos) or MFCC features (from audio files) and prepares the input data for analysis.



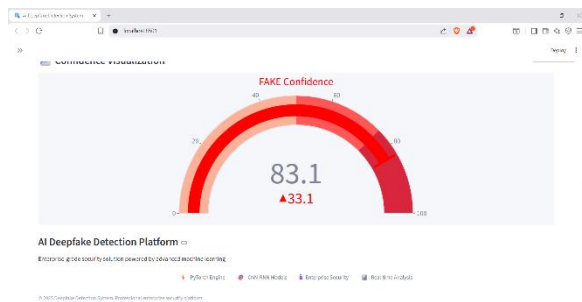
4) The CNN/GRU model performs spatial and temporal feature extractions on the preprocessed input data.



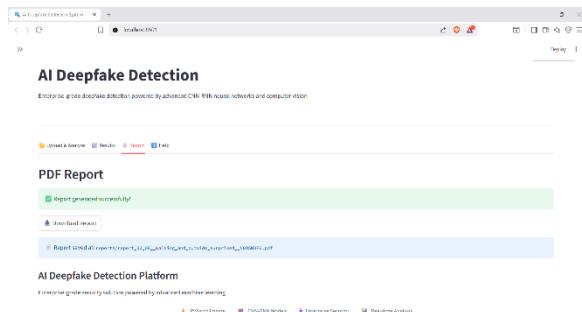
5) The attention mechanism identifies areas of concern within the uploaded data, thus improving the decision-making process of the model.



6) The classifier will provide either a prediction as to whether or not the input data is real (true) or fabricated (false), along with an associated confidence score.



7) Finally, the system will either return a result or generate a downloadable PDF report.



VI. Conclusion and Future Work

In our paper, we have shared our journey of secretly developing a new deepfake detector. More importantly, we have demonstrated a couple of cool features such as the unified audio, video component and the real, world, friendliness without making deepfake detectors the only focus. We implemented a combination of Convolutional Neural Networks (CNNs) to extract spatial cues, recurrent layers (BiLSTM) to capture temporal dependencies, and attention mechanisms to concentrate on the most telling frames, which greatly improves the performance of the model compared to the methods only using facial frames. Moreover, the full, stack implementation with secure handling, detailed logging, and automatic forensic reports converts a mere laboratory prototype into a feasible solution that can be integrated into cybersecurity pipelines, forensic laboratories, or media verification workflows.

The experiments pretty much confirmed this. On well, known datasets such as FaceForensics++, Celeb, DF

and DFDC, our method consistently improved detection performance and, most importantly, yielded greater resilience to various degradations that are typical of the real, world conditions: severe compression, noise, and low resolution. Such sturdiness plus the multimodality feature altogether evoke the notion of a significant breakthrough in the arsenal of combating the attacks where the adversaries are hard to spot because of their trickery in the visuals as well as in the auditory fakes.

However, it really isn't the story's end. In fact, several avenues of the project are worth exploring.

Replacing the current transformer blocks with lighter versions (or even hybrid recurrent, transformer blocks) might be a way to increase the length of dependencies without drastically increasing the inference time. Ensuring that the audio and video tell compatible stories thus cross, modal consistency seems like a ripe avenue for detecting even more sophisticated attacks.

Perhaps, adversarial training would almost certainly be a weapon against evasion attempts, and adding better explainability (beyond just attention maps) would make outputs easier for non, experts to trust and audit.

More generally, we want to delve into federated learning configurations to train on decentralized, privacy, preserving data sources, to continue to overcoming dataset biases through more diverse collection and augmentation, and to figure out the whole pipeline to work for an edge device or large, scale cloud deployment.

The big picture here is the creation of non, static defenses, systems that are able to adjust when new generative models (diffusion, based, multimodal large models, etc.) keep being released. It is definitely a race for superiority, but with smart engineering and maintaining the collaboration across the different fields, we can be the ones leading instead of always being one step behind.

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